HADROS

HADRONIC ASTROPHYSICS FOR DIS, RADIATION AND OPAQUE SYSTEMS

HADROS User Manual

High-energy Astrophysical DIS Radiation and Observation Simulator

HADROS

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Contents

1 Overview

HADROS combines Kerr ray tracing, semi-analytic astrophysical backgrounds, UHE neutrino DIS opacity, phenomenological source and spectral prescriptions, diagnostic MeV neutrino post-processing, ParaView export, and a static dashboard for organizing plots.

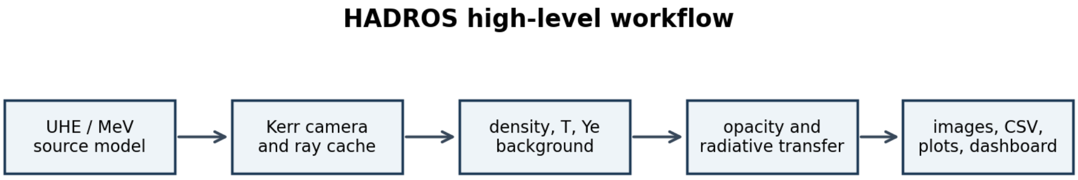

Figure 1: High-level HADROS workflow.

At a high level the pipeline is

$$\text{source} \to \text{Kerr rays} \to \text{background} \to \text{opacity/transfer} \to \text{images and diagnostics.}$$

HADROS does not perform hydrodynamic evolution. It is not full neutrino radiation hydrodynamics, not a calibrated collapsar simulation, and not a complete PDF-uncertainty package. The current backgrounds are semi-analytic density morphologies, the UHE sources are phenomenological prescriptions, and the MeV luminosities are diagnostic proxies.

2 Physical Theories

2.1 Kerr Ray Tracing

The spacetime around a rotating black hole is modeled with the Kerr metric. The main physical parameters are the black-hole mass `MBH_MSUN` and the dimensionless spin `ASPIN`. The observer is placed at a large radius `CAM_R_OBS_RG` and inclination `CAM_THETA_DEG`. The camera launches rays backward through the Kerr geometry and stores geodesics in a cache.

For neutrinos in the UHE range, the rest mass is negligible compared with the energy, so the propagation is approximated by null geodesics:

$$ds^2 = 0.$$

The ray path is therefore controlled by the Kerr geometry. The energy dependence in UHE images mostly enters through opacity, $\sigma_{\nu N}(E)$, not through chromatic lensing.

HADROS works in gravitational-radius units,

$$r_g = \frac{GM_{\rm BH}}{c^2}.$$

Radii, camera distances, source locations, and ray steps are usually expressed in units of r_g. The black-hole mass enters again when converting a dimensionless path length into centimeters for opacity integrals.

Spin changes more than the horizon size. In Kerr spacetime frame dragging modifies the mapping between camera pixels and paths near the black hole. The outer horizon is

$$r_+ = 1 + \sqrt{1 - a^2}$$

in geometric units. HADROS treats the spin as a controlled input parameter, not as the result of a self-consistent accretion calculation.

The ray cache should be understood as a map from image pixels to paths through the environment. Pixels near the shadow correspond to rays that pass close to the horizon or are captured; pixels away from the shadow sample more direct paths through the torus and funnel.

2.1.1 Equations actually solved by HADROS

The Kerr module is implemented in `src/kerr_metric.cpp`, `src/kerr_geodesic.cpp`, and `src/kerr_camera.cpp`. It uses Boyer-Lindquist coordinates (t, r, θ, ϕ) with signature $(-, +, +, +)$ and dimensionless units $G = M = c = 1$. The code defines

$$\Sigma = r^2 + a^2 \cos^2\theta, \qquad \Delta = r^2 - 2r + a^2,$$
$$A = (r^2 + a^2)^2 - a^2\Delta \sin^2\theta.$$

The nonzero covariant metric components are

$$g_{tt} = -\left(1 - \frac{2r}{\Sigma}\right), \qquad g_{t\phi} = -\frac{2ar\sin^2\theta}{\Sigma},$$
$$g_{rr} = \frac{\Sigma}{\Delta}, \qquad g_{\theta\theta} = \Sigma,$$
$$g_{\phi\phi} = \left(r^2 + a^2 + \frac{2a^2 r\sin^2\theta}{\Sigma}\right)\sin^2\theta.$$

The inverse metric used by the geodesic Hamiltonian is

$$g^{tt} = -\frac{A}{\Sigma\Delta}, \qquad g^{t\phi} = -\frac{2ar}{\Sigma\Delta},$$
$$g^{rr} = \frac{\Delta}{\Sigma}, \qquad g^{\theta\theta} = \frac{1}{\Sigma},$$
$$g^{\phi\phi} = \frac{\Delta - a^2\sin^2\theta}{\Sigma\Delta\sin^2\theta}.$$

Null geodesics are evolved in Hamiltonian form. The state vector is

$$y = (t, r, \theta, \phi, p_t, p_r, p_\theta, p_\phi),$$

where the momenta are covariant. The Hamiltonian is

$$H = \frac{1}{2} g^{\mu\nu} p_\mu p_\nu,$$

and a null ray satisfies $H = 0$ up to numerical error. The canonical equations implemented in the code are

$$\frac{dx^\mu}{d\lambda} = \frac{\partial H}{\partial p_\mu} = g^{\mu\nu} p_\nu,$$
$$\frac{dp_\mu}{d\lambda} = -\frac{\partial H}{\partial x^\mu} = -\frac{1}{2}\partial_\mu g^{\alpha\beta} p_\alpha p_\beta.$$

Stationarity and axisymmetry imply

$$\frac{dp_t}{d\lambda} = 0, \qquad \frac{dp_\phi}{d\lambda} = 0,$$

so the code evolves p_r and p_θ using derivatives of the inverse metric. These derivatives are computed by centered finite differences:

$$\partial_r g^{\mu\nu} \simeq \frac{g^{\mu\nu}(r+\epsilon, \theta) - g^{\mu\nu}(r-\epsilon, \theta)}{2\epsilon},$$

$$\partial_\theta g^{\mu\nu} \simeq \frac{g^{\mu\nu}(r,\theta+\epsilon) - g^{\mu\nu}(r,\theta-\epsilon)}{2\epsilon}.$$

The current code uses $\epsilon = 10^{-5}\max(1,|r|)$ for the radial derivative and $\epsilon = 10^{-5}$ for the angular derivative.

The camera initializes each pixel in a local ZAMO/LNRF-like frame. Pixel coordinates are mapped to local angular camera coordinates

$$u = \left[\frac{2(i+0.5)}{N_x} - 1\right]\tan\frac{\mathrm{FOV}}{2}, \qquad v = \left[\frac{2(j+0.5)}{N_y} - 1\right]\tan\frac{\mathrm{FOV}}{2},$$

$$\mathcal{N} = \sqrt{1+u^2+v^2}.$$

The local backward ray direction is

$$n_r = -\frac{1}{\mathcal{N}}, \qquad n_\theta = \frac{v}{\mathcal{N}}, \qquad n_\phi = \frac{u}{\mathcal{N}}.$$

Using the lapse α, frame-dragging angular velocity ω, and spatial metric components, the code sets

$$p^t = \frac{1}{\alpha}, \qquad p^r = \frac{n_r}{\sqrt{g_{rr}}}, \qquad p^\theta = \frac{n_\theta}{\sqrt{g_{\theta\theta}}},$$

$$p^\phi = \frac{n_\phi}{\sqrt{g_{\phi\phi}}} + \omega p^t.$$

It then lowers the index with $p_\mu = g_{\mu\nu}p^\nu$ before integration.

2.1.2 Numerical integration method

The production geodesic stepper is an adaptive Runge-Kutta-Fehlberg 4(5) method, implemented as `KerrGeodesic::step_adaptive`. It computes both a fourth-order estimate y_4 and a fifth-order estimate y_5 using the classic Fehlberg stages:

$$k_1 = f(y),$$

$$k_2 = f\left(y + \frac{h}{4}k_1\right),$$

$$k_3 = f\left(y + h\left[\frac{3}{32}k_1 + \frac{9}{32}k_2\right]\right),$$

$$k_4 = f\left(y + h\left[\frac{1932}{2197}k_1 - \frac{7200}{2197}k_2 + \frac{7296}{2197}k_3\right]\right),$$

$$k_5 = f\left(y + h\left[\frac{439}{216}k_1 - 8k_2 + \frac{3680}{513}k_3 - \frac{845}{4104}k_4\right]\right),$$

$$k_6 = f\left(y + h\left[-\frac{8}{27}k_1 + 2k_2 - \frac{3544}{2565}k_3 + \frac{1859}{4104}k_4 - \frac{11}{40}k_5\right]\right).$$

The error estimate is the maximum absolute difference between selected components of y_4 and y_5:

$$\epsilon_{\mathrm{err}} = \max(|r_4 - r_5|, |\theta_4 - \theta_5|, |\phi_4 - \phi_5|, |p_{r,4} - p_{r,5}|, |p_{\theta,4} - p_{\theta,5}|, |p_{\phi,4} - p_{\phi,5}|).$$

If this error is below the tolerance, the fifth-order result is accepted. If not, the step is retried with a smaller step. The update factor is

$$q = 0.8\left[\frac{\mathrm{tolerance}}{\max(\epsilon_{\mathrm{err}}, 10^{-30})}\right]^{1/4},$$

clamped between 0.1 and 5.0. The minimum internal step is $h_{\min} = 10^{-5}h$. If the adaptive routine fails after 50 attempts, HADROS falls back to a fixed fourth-order Runge-Kutta step.

A ray is marked as captured when

$$r \leq r_+ + 10^{-3}.$$

It exits the computational domain when $r \geq r_{\max}$ after the initial transient. A hard cap of 200000 steps prevents accidental infinite integrations. For each stored path point, HADROS records

$$(r, \theta, x, y, z, dl, \text{redshift factor}),$$

where dl is estimated from the spatial Boyer-Lindquist metric components at the midpoint between consecutive stored points. The resulting ray cache is then reused by the opacity and radiative-transfer routines.

2.2 Semi-analytic Density Backgrounds

The matter field is a controlled semi-analytic background, not a hydrodynamic solution. Available density profiles include `gaussian`, `powerlaw`, funnel variants, envelope variants, and `collapsar_ndaf_like`.

Every profile applies an explicit density floor:

$$\rho = \max(\rho_{\text{raw}}, \rho_{\text{floor}}).$$

The profiles are deliberately simple enough to scan. A Gaussian torus is useful as a control; a power-law torus gives a more extended radial column; funnel variants reduce polar density; and envelope variants test whether outer collapsar-like material contributes appreciably to UHE attenuation.

These profiles are morphology, not dynamics. There is no pressure balance equation, angular-momentum transport, self-consistent heating, or hydrodynamic time evolution. The advantage is interpretability: changing one parameter changes a controlled column-density geometry.

The temperature and electron-fraction profiles used by the MeV module are also parameterized. A MeV luminosity computed from these fields is a diagnostic consequence of the assumed semi-analytic background, not a calibrated simulation prediction.

Regime	Purpose	Interpretation and limitation
`fiducial_uhe_default`	UHE baseline	Low-density transparent test background; not a collapsar disk.
`fiducial_mev_density`	MeV baseline	Around 10^9–10^{10} g cm^{-3} in audits; weakly cooled torus with expected low luminosity.
`collapsar_ndaf_like`	high-density preset	Significant regions near 10^{10}–10^{12} g cm^{-3}; semi-analytic NDAF/collapsar-like regime, not hydrodynamics.

2.3 UHE Neutrino DIS Opacity

The UHE optical depth is computed from the baryon column along a ray:

$$\tau_{\text{DIS}}(E) = \int n_b(s)\, \sigma_{\nu N}(E)\, ds, \qquad P_{\text{surv}}(E) = \exp[-\tau_{\text{DIS}}(E)].$$

HADROS currently compares `GBW`, `IIM`, `CTW_reference`, and `literature_powerlaw_scale`. `CTW_reference` is the central νN charged-current table from Connolly, Thorne & Waters, Phys. Rev. D 83, 113009, Table I. It does not include neutral-current interactions, antineutrino tables, regeneration, or PDF uncertainty bands.

The opacity integral separates the astrophysics from the particle physics:

$$\tau_{\rm DIS} \sim \text{target column} \times \text{interaction probability}.$$

The target column is set by $n_b ds$ and the cross section is set by $\sigma_{\nu N}(E)$. This separation allows the same ray cache and density field to be post-processed with GBW, IIM, CTW, or another future table.

If $\tau \ll 1$, then $P_{\rm surv} \simeq 1 - \tau$ and DIS-model differences produce small brightness changes. If $\tau \gg 1$, all models may give zero observed intensity even if their absolute optical depths differ by large factors. For this reason the validation reports compare both optical-depth metrics and image intensity.

At UHE energies the cross section probes very small Bjorken x. GBW and IIM are saturation-inspired small-x models, while CTW is a Standard-Model PDF-based reference using MSTW 2008 PDFs. HADROS does not decide which model is correct; it propagates each assumption through the same astrophysical setup.

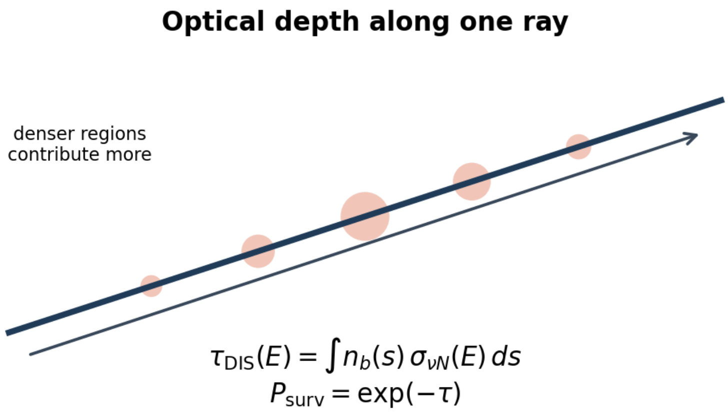

Figure 2: Optical depth and survival probability along a ray.

2.4 UHE Source Morphologies

The UHE emissivity is separated into a source morphology and a spectral model. Implemented source morphologies are `inner_ring`, `funnel_wall`, `jet_base`, `shock_layer`, and `density_weighted`. These are phenomenological source morphologies, not first-principles particle-acceleration simulations.

The source morphology changes where neutrinos are injected, not how the medium attenuates them. Equatorial injection tends to cross larger baryonic columns; funnel-wall injection can expose lower-density escape paths; jet-base injection emphasizes polar geometry and strong lensing. The `shock_layer` option is a density-gradient proxy, not a shock-acceleration calculation.

2.5 UHE Spectral Models

The spectral interface currently supports `monochromatic`, `powerlaw`, and `powerlaw_cutoff`. The power-law spectrum is

$$\frac{dN}{dE} \propto E^{-\gamma},$$

and the cutoff model is

$$\frac{dN}{dE} \propto E^{-\gamma} \exp(-E/E_{\rm cut}).$$

Energy-band composite images use false colors: blue for low-energy UHE, green for intermediate-energy UHE, and red for high-energy UHE.

A monochromatic image asks what the sky looks like at one energy. A spectral image weights several energies by an emitted spectrum and then attenuates each energy differently. Since $\sigma_{\nu N}(E)$ grows with energy, the observed spectrum can be softer than the emitted spectrum. The spectral interface is generic: future spectra can be added by changing the spectral-weight function, not the ray-tracing or opacity pipeline.

2.6 Optical-depth Surfaces

HADROS can extract diagnostic opacity surfaces such as $\tau = 2/3$, $\tau = 1$, and $\tau = 3$. These surfaces are properties of the medium, geometry, energy, and DIS cross section. They do not depend on the UHE source morphology. The current extraction is axisymmetric and returns $r_\tau(\theta)$.

An opacity surface is analogous in spirit to a photosphere, but it is not an emission surface. A surface $\tau = 1$ means material inside that radius is attenuating according to the chosen convention. For UHE neutrinos this is a DIS opacity surface, not a thermal neutrinosphere. Since it is a property of the medium, it should remain unchanged when only the UHE source morphology changes.

2.7 Thermal MeV Neutrino Module

The MeV module is separate from UHE/DIS. It is a diagnostic local emissivity and radiative-transfer model, not a calibrated absolute luminosity prediction. Implemented channels include URCA-like processes,

$$e^- + p \to n + \nu_e, \qquad e^+ + n \to p + \bar{\nu}_e,$$

pair annihilation,

$$e^- + e^+ \to \nu + \bar{\nu},$$

and nucleon-nucleon bremsstrahlung,

$$N + N \to N + N + \nu + \bar{\nu}.$$

The local transfer equation is

$$\frac{dI}{ds} = j - \alpha I.$$

CPU MeV is the canonical physical reference implementation. CUDA MeV remains legacy/toy until it is ported and validated.

The MeV and UHE modules live in different physical regimes. MeV neutrino emission is thermal or weak-interaction emission from hot dense matter; UHE attenuation is DIS scattering on baryons at vastly higher energies. They can be evaluated on the same background, but they should not be mixed conceptually.

Temperature is crucial. Pair emission rises very steeply with temperature, often summarized by high-power scalings such as T^9 in simple arguments. URCA-like channels also rise rapidly and depend on density and electron fraction. Thus a torus at $T \sim 4$ MeV can be orders of magnitude dimmer than a torus at $T \sim 15$ MeV.

Electron fraction Y_e controls the neutron/proton balance and therefore the relative behavior of ν_e, $\bar{\nu}_e$, and heavy-lepton flavors grouped as ν_x. In the current approximation, ν_x does not have the same charged-current absorption as electron-flavor neutrinos.

The transfer equation has useful limits. In the optically thin case,

$$I_{\rm out} \simeq I_{\rm in} + j\, ds,$$

while for large αds the intensity approaches the source function,

$$I_{\text{out}} \to \frac{j}{\alpha}.$$

The CPU MeV validation checks these limiting behaviors.

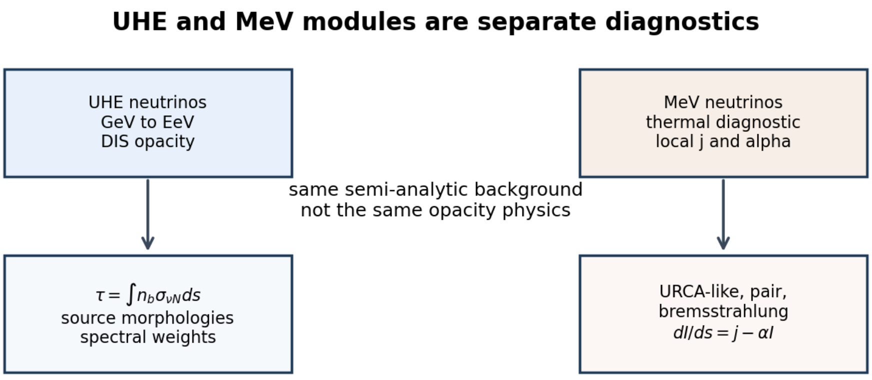

Figure 3: UHE and MeV modules are separate diagnostics.

2.8 MeV Diagnostic Upgrades

The MeV workflow includes temperature profiles, electron-fraction profiles, Fermi-Dirac-like spectral integration, MeV neutrinosphere extraction at $\tau_{\text{MeV}} = 2/3$, electron-degeneracy diagnostics, opacity-component decomposition, and luminosity-proxy audits.

2.9 ParaView and 3D Visualization

The ParaView export samples the current axisymmetric semi-analytic model onto a 3D Cartesian grid. It is not a fully 3D hydrodynamical simulation and does not export a fake local optical-depth field.

3 Main Makefile Commands

3.1 Safe Build and Validation

```
make
make build NTHREADS=2
make validate_small NTHREADS=2
make validate_small_cuda NTHREADS=2 \
  NVCC=/home/rafael/micromamba/envs/dis/bin/nvcc \
  PYTHON=/home/rafael/micromamba/envs/dis/bin/python
make validate_medium_cuda NTHREADS=2 \
  NVCC=/home/rafael/micromamba/envs/dis/bin/nvcc \
  PYTHON=/home/rafael/micromamba/envs/dis/bin/python
```

3.2 Running Images

```
make image-from-cache ENU=1e11 DENSITY_PROFILE=gaussian NTHREADS=2
```

```
make image-from-cache SPECTRAL_MODEL=powerlaw \
  SPECTRAL_GAMMA=2.0 SPECTRAL_E_MIN_GEV=1e5 \
  SPECTRAL_E_MAX_GEV=1e12 SPECTRAL_N_BINS=8 NTHREADS=2
make plot_multiband_image PYTHON=python3
make image-from-cache MEV_MODEL=physical MEV_FLAVOR=anti_nu_e NTHREADS=2
make mev_multiband_image NTHREADS=2 PYTHON=python3
```

3.3 Robustness, DIS, Opacity, and MeV Diagnostics

```
make robustness_scans NTHREADS=2 PYTHON=python3
make validate_dis_models NTHREADS=2 PYTHON=python3
make opacity_surfaces TAU_SURFACE_VALUE=1.0 PYTHON=python3
make validate_mev_physics PYTHON=python3
make mev_neutrinosphere PYTHON=python3
make mev_luminosity PYTHON=python3
make audit_mev_luminosity PYTHON=python3
make audit_torus_regime PYTHON=python3
make audit_collapsar_ndaf_like PYTHON=python3
```

3.4 ParaView and Dashboard

```
make paraview_fields \
  DENSITY_PROFILE=powerlaw_funnel_envelope \
  SOURCE_MODEL=funnel_wall \
  PARAVIEW_NX=64 PARAVIEW_NY=64 PARAVIEW_NZ=64 \
  PARAVIEW_BOX_RG=80 NTHREADS=4

make dashboard PYTHON=python3
```

Open `dashboard/index.html` in a browser. The dashboard indexes existing plots and output products; it does not run simulations.

3.5 Thread Control

Use explicit limits:

```
make -j2 build
make validate_small NTHREADS=2
make run_production NTHREADS=4
```

The default `make` target prints help and does not launch a production simulation.

4 Outputs and Directory Map

- `data/`: input data tables, especially DIS sigma tables.
- `include/`: C++ headers.
- `src/`: core implementations.
- `apps/`: command-line applications.
- `scripts/`: plotting, validation, and audit scripts.
- `docs/`: manuals and scientific notes.
- `plots/`: generated figures.
- `output/`: generated tables, images, ray caches, and reports.
- `dashboard/`: static HTML dashboard.

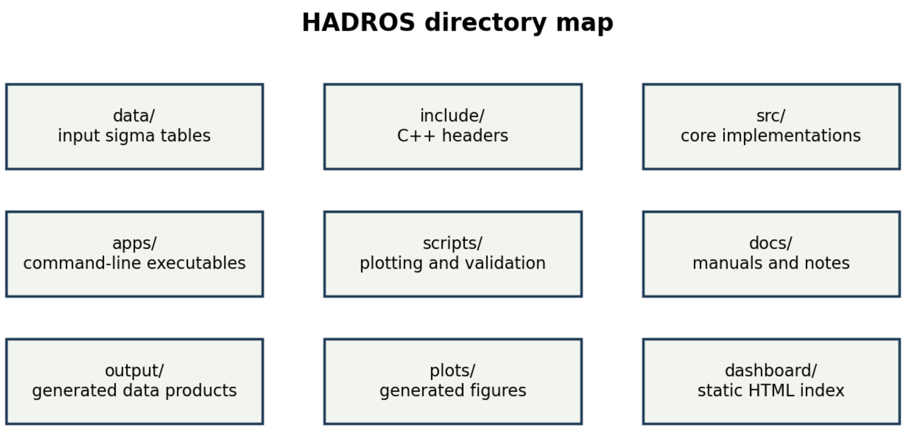

Figure 4: Main HADROS directories.

5 Program Workflow

The internal workflow is:

1. choose a physical setup;
2. generate or load a Kerr ray cache;
3. evaluate the local background along each ray;
4. compute local UHE and/or MeV emissivity;
5. compute opacity;
6. integrate radiative transfer along rays;
7. write data tables;
8. generate plots;
9. index products in the dashboard.

6 Recommended Workflows for Students

Beginner dashboard:

```
make dashboard PYTHON=python3
```

Beginner validation:

```
make validate_small NTHREADS=2 PYTHON=python3
```

Intermediate DIS comparison:

```
make validate_dis_models NTHREADS=2 PYTHON=python3
```

Intermediate MeV diagnostics:

```
make validate_mev_physics PYTHON=python3
```

Advanced collapsar/NDAF-like comparison:

```
make audit_collapsar_ndaf_like PYTHON=python3
```

Advanced ParaView export:

```
make paraview_fields DENSITY_PROFILE=collapsar_ndaf_like NTHREADS=4
```

7 Scientific Limitations

- Backgrounds are semi-analytic, not hydrodynamic evolutions.
- HADROS does not solve full neutrino radiation hydrodynamics.
- MeV luminosities are proxy/diagnostic quantities.
- CUDA MeV remains legacy/toy and is not equivalent to CPU physical MeV.
- `CTW_reference` is central charged-current νN only.
- UHE neutral-current interactions and regeneration are not yet included.
- There is no true 3D hydrodynamic background yet.

8 Troubleshooting

If `make` seems to run too long, stop the process and use a validation target such as `validate_small`. If CPU usage is too high, pass `NTHREADS=2` or `NTHREADS=4`, and use limited make parallelism such as `make -j2 build`. If CUDA is not found, check `which nvcc`, `nvcc -version`, and `nvidia-smi`. If plots are missing, regenerate the relevant plot target and rebuild the dashboard. If stale caches are suspected after a crash, see `README_RECOVERY_20260609.md`.

9 Glossary

DIS Deep inelastic scattering, the high-energy neutrino-nucleon interaction used for UHE opacity.

UHE Ultra-high energy, here referring to GeV-to-EeV scale neutrinos.

MeV Thermal neutrino energy scale used in the diagnostic module.

Optical depth Dimensionless attenuation measure, τ.

Survival probability $P_{\text{surv}} = \exp(-\tau)$.

Neutrinosphere Diagnostic surface where optical depth reaches a chosen value, often $\tau = 2/3$.

Kerr metric Spacetime geometry around a rotating black hole.

NDAF Neutrino-dominated accretion flow.

Collapsar Massive-star collapse scenario that can form a black hole and dense accretion environment.

CTW Connolly, Thorne & Waters reference cross-section table.

GBW Golec-Biernat-Wusthoff saturation-inspired DIS model.

IIM Iancu-Itakura-Munier saturation-inspired DIS model.